

Hands-on Coding Challenge (Rust + SQL + Containers)

Build a minimal **Rust microservice** (only backend) with:

Requirements:

1. **Rust API endpoint** /certificates that:
 - Accepts JSON input for certificate metadata
 - Inserts data into **PostgreSQL**
 - Retrieves metadata on GET /certificates/:id
2. **PostgreSQL schema** for:
 - Certificate ID
 - Subject
 - Expiration
 - Issuer
 - SAN entries (array or secondary table)
3. **Dockerfile** to build and run the service
4. Bonus (strong differentiator):
 - Implementing a certificate parser in Rust
 - Using an async Rust framework (Axum, Actix, or Warp)
 - Using SQLx or Diesel with compile-time checks
 - Writing unit tests
 - Container security best practices (non-root user, slim base image)